

Proposal for Second Year Paper

Ke Jiang

1 Introduction

1.1 Motivation

U.S cities on average show lower transit infrastructure, Freemark, 2023 reported that even the transit length and ridership are growing globally, the United States seems to be stagnated since 1980s. Putting aside China, whom contribute to 40% of total metro length, by 2023 the total metro length of United States was surpassed by India and nearly half of EU(fig 2). Even incorporating light rails or streetcars, which U.S cities put more focus on, the construction of transit still declines since 1970s and struggles to catch up the population growth. Reasons behind this can be puzzling but nontrivial. Transportation infrastructures are often endogeneous, it serves as a tool to boost economic development by reducing spatial frictions but can also be distorted(Donaldson, 2025). In particular, policymakers may intentionally build the infrastructure to boost economics, or they may choose not to build if it's too expensive compared to returns. In general they are involved in scenario where they evaluate the returns and costs of new infrastructure. Standard evaluation of the returns to transportation infrastructure involves measuring the travel time saved or more distributional effect(Fogel, 1964; Tsivanidis, 2023; Severen, 2021). Both Tsivanidis (2023) and Severen (2021) find that the returns to new transit infrastructure are lower than expected, and even negative from Severen (2021), due to the limited housing supply near stations. Severen (2021) also pointed out the dispersed workplaces and polycentric city structure as another reason for the low return to transit. In summary restrictive zoning can limit the housing supply response to transit investment, which in turn potentially discourage transit investment.

However, literature on estimating transportation infrastructure often take these housing-supply restrictions as given, whereas in reality, aside from certain geographic constraints, such restrictions are often the result of local zoning policies. Zoning is often politically supported by incumbent residents because it protects neighborhood quality and property values. In particular, restrictive zoning can limit housing supply, which protects local property values, reduces congestion in local public goods and amenities, and preserves environmental quality and neighborhood character(Rollet and Weiwu, 2025). However, these local benefits may come at broader citywide costs. In particular, restrictive zoning reduces the elasticity of housing supply in desirable locations(Glaeser and Gyourko, 2002), and land-use regulation pushes forward job-housing separation, raises commuting costs, and contributes to urban sprawl(Glaeser and Kahn, 2003). But the low-density city layout can also lower the economic return to public transit and discourage investment in public transit. Lower transit investment can turn into more car-oriented commuting, which further decentralize the city(Baum-Snow, 2007). Such decentralization and car-dominance in turn reinforce the demand for low-density city and therefore the zoning regulations.

This self-enforcing pattern can be a key driver of urban sprawl, and may help explain the stagnation of transit infrastructure in U.S cities.

This self-reinforcing pattern between restrictive zoning and low transit investment can be particularly problematic in the context of U.S cities. On the one hand, low-density sprawl requires more road infrastructures per capita, the cost and maintenance fees may not be sustainable (Brueckner, 2000). On the other hand, the car-dependence and long commuting distance can increase congestion which are not internalized by households. Moreover, the low transit investment can also limit the accessibility of low-income households to job opportunities, which can exacerbate economic inequality (Kawano2023-ly).

In summary, the discussion points to a broader but maybe underexplored problem: the full effect of zoning could be potentially more than what we currently understand, if we consider its effect on endogenous transit investment decisions. In particular, this proposal aims to explore the following questions: First, how does local zoning affect metropolitan transportation investment decisions? Second, what are the welfare implications of this interaction, and how much inefficiency can be mitigated compared to a socially optimal benchmark where zoning and transportation are coordinated?

1.2 Literature Review

This proposal draws on three interconnected bodies of literature: studies on transportation infrastructure and its evaluation, literature on zoning and land-use regulation, and research on urban sprawl and city growth patterns.

1.2.1 Transportation Infrastructure, Optimal Structure, and Evaluation

A large body of literature examines how transportation infrastructure shapes spatial economic patterns and how to evaluate infrastructure investments. Donaldson (2025) provides a comprehensive framework for evaluating transportation infrastructure impacts, emphasizing the importance of general equilibrium effects and showing how infrastructure changes affect economic activity through multiple channels. The quantitative spatial economics framework is essential for this evaluation: Redding (2025) surveys quantitative urban economics methods, while Redding and Rossi-Hansberg (2017) details the development of quantitative spatial models that can integrate multiple dimensions of infrastructure impacts. Contemporary applications of these methods highlight that traditional benefit-cost analyses may not capture full infrastructure impacts. Severen (2026) describes how quantitative spatial models can evaluate transportation improvements within cities, providing practical guidance for implementation. Importantly, Fajgelbaum et al. (2021) develops a framework integrating spatial epidemiology and commuting networks, demonstrating the importance of accounting for spatial structure in policy design. The returns to transit infrastructure specifically have received attention. Tsivanidis (2023) evaluates Bogotá’s TransMilenio system and finds that transit returns depend critically on complementary policies affecting land use. Baum-Snow and Kahn (2000) studies rail transit expansion in five U.S. cities and finds modest impacts on usage and housing values, suggesting limitations to transit investment absent other changes. Kahn (2007) documents significant heterogeneity in transit effects across 14 cities, with gentrification outcomes varying substantially based on local conditions like station type (walk-and-ride versus park-and-ride).

The interaction between transportation infrastructure and congestion is increasingly recognized. Akbar et al. (2023) analyzes the relationship between mobility and conges-

tion in urban India, showing how congestion affects the effectiveness of transportation improvements. Allen and Arkolakis (2022) develops a general equilibrium framework with endogenous congestion to evaluate welfare impacts of transportation improvements, finding substantial variation in returns across different infrastructure investments. Kashner and Ross (2025) relates congestion to upzoning, showing that the effect of upzoning on housing supply depends on where upzoning occurs. Specifically, upzoning near transit stations has lower congestion effects than upzoning in other areas. Integrating congestion into the analysis of both zoning and transportation is critical for my proposal, which I believe may be a key channel explaining welfare differences. Still my proposal contribute by modeling the endogeneity of zoning regulations and transportation, which is often taken as given in the literature.

1.2.2 Zoning and Land-Use Regulation

Zoning and land-use regulations have emerged as critical determinants of urban form and economic efficiency. The foundational work of Glaeser and Gyourko (2002) demonstrates that zoning restrictions significantly reduce housing supply elasticity, with particularly severe constraints in high-demand cities. This finding has profound implications: Hsieh and Moretti (2019) quantifies the aggregate costs of housing constraints, estimating that zoning restrictions lowered U.S. aggregate growth by 36 percent between 1964 and 2009 by preventing spatial reallocation of labor to high-productivity locations.

Recent work has continued this examination with new evidence and methods. Rollet and Weiwu (2025) measures winners and losers from increasing housing supply, highlighting the distributional tensions inherent in zoning reform—local residents benefit from restrictive zoning even when it imposes broader costs. Song (2025) estimates the effects of residential zoning on U.S. housing markets, providing detailed evidence of how zoning shapes housing supply responses. Mast (2024) uses electoral reforms as a source of exogenous variation in local control and finds that decentralization of land-use authority reduces housing permits by 20 percent, with larger effects in affluent areas. These results highlight how local political economy affects housing supply. Importantly, Baum-Snow and Duranton (2025) surveys housing supply and affordability, providing context for understanding how zoning interacts with other factors affecting housing markets. A critical insight from recent work is that land-use regulations interact with transportation investment. Hu and Wang (2025) studies how subway expansion affects outcomes under different zoning regimes, finding that rigid land-use regulation limits the benefits of transit infrastructure. My proposal tries to emphasize that restrictive zoning also discourages transit investment, which is a key channel that has not been fully explored in the literature.

1.2.3 Urban Sprawl and City Growth

Urban sprawl represents a fundamental concern in metropolitan economics. Brueckner (2000) provides the canonical analysis of sprawl causes and remedies, identifying three market failures: incomplete internalization of open-space benefits, failure to charge commuters for congestion externalities, and insufficient cost-recovery for infrastructure. Baum-Snow (2007), discussed above, documents empirically how highways have driven suburban decentralization, with implications for sprawl patterns. Glaeser and Kahn (2003) documents that sprawl is ubiquitous and continuing to expand, but is not caused by bad urban planning; rather, it reflects increasing demand for car-based living, though

low-income people who cannot afford cars are left behind. My proposal seeks to explain another mechanism where car ownership may not merely be a preference but also an equilibrium outcome.

The literature collectively demonstrates that sprawl outcomes reflect a complex interaction between fundamental forces, such as population growth, rising incomes, falling transport costs, and structural changes. Cuberes, Desmet, and Rappaport (2021) documents urban growth shadows and access effects, showing how proximity to large metropolitan areas affects growth patterns through both competition and market-access mechanisms. Desmet and Rappaport (2017) traces the long-term evolution of U.S. settlement patterns from 1800-2000, documenting the transition toward increasingly dispersed city systems. Structural change and land use are intimately connected. Coeurdacier, Oswald, and Teignier (2022) studies how structural change in the economy drives land-use changes and urban expansion. International evidence adds important insights: Zheng and Kahn (2013) documents how China’s bullet trains facilitate market integration and mitigate megacity growth costs by allowing workers to access large labor markets from secondary cities. Building on these intergrated insights, my proposal aims to explore how the interaction between zoning and transportation investment shapes sprawl patterns, and how more coordinated policies could mitigate sprawl and its associated costs.

2 Model

Following Severen (2026), I aim to develop a quantitative spatial model to answer the questions raised in the introduction. The model will capture the key features of metropolitan spatial structure, including heterogeneous locations, commuting patterns, and land-use decisions. It will also incorporate the institutional structure of local zoning and metropolitan transportation investment. Before delving into the full model, I first present a simplified toy model to illustrate the key mechanisms at play. (In progress and may be revised in the future)

2.1 Toy model

For simplicity, consider a monocentric model where every household commutes to the city center (CBD) to work, earning a uniform wage w , net of commuting cost τx , where $\tau = \tau(N, I)$ is the unit travel cost subject to travel volume and transportation investment. First, let’s assume the economy is closed, such that total population $\bar{N}$ is fixed. Households arbitrage between commuting costs and housing units. There is a local municipality responsible for area $x \in (0, x_m]$, while for area (x_m, x_f) there are unregulated areas, and x_f determines the city fringe.

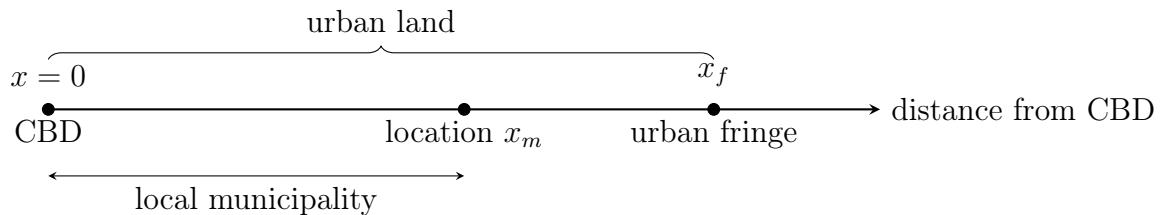


Figure 1: A line-segment representation of the monocentric city model

2.1.1 Household

Following Glaeser and Kahn, 2003, I write household utility as follows:

$$u(c, h) = c + \alpha \log h \quad (1)$$

with budget constraint $w - \tau x = c + hp(x)$, where $p(x)$ is the housing price. Optimal allocation of housing gives $p(x)h = \alpha$. Equilibrium implies households are indifferent across locations, such that $p'(x)h + \tau = 0$. These two conditions pin down the housing price in terms of location x :

$$P(x) = P(0) \exp\left(-\frac{\tau}{\alpha}x\right) \quad (2)$$

2.1.2 Developer

Housing supply on each parcel of land is provided by competitive developers who solve:

$$\max \pi = pS^\gamma - S - r \quad (3)$$

where S is structural density (housing capital per land unit), which captures housing intensity, or building height. By setting $\gamma \in (0, 1)$, housing production exhibits decreasing returns to scale. In the competitive, free-entry developer market, $r(x) = p(x)S(x)^\gamma - S(x)$, which pins down the land price. Solving the developer's problem gives $s(x) = (\gamma p(x))^{1/(1-\gamma)}$. Combining housing supply and housing demand, population density is determined by:

$$n(x) = \frac{H(x)}{h(x)} = \frac{p(x)S(x)^\gamma}{\alpha} \quad (4)$$

2.1.3 Local Municipality

The local municipality responsible for location $(0, x_m]$ cares about land prices and local quality of life, which is negatively correlated with density. I formulate the local municipality's problem as follows:

$$\max_{s(x)} \int_0^{x_m} [r(x) - \omega n(x)] dx \quad (5)$$

where ω captures the marginal disutility from density. Combined with previous conditions, this can be refined as:

$$\max_{s(x)} \int_0^{x_m} p(x)S(x)^\gamma - S(x) - \omega \frac{p(x)S(x)^\gamma}{\alpha} dx \quad (6)$$

The optimal structural density is $S_m(x) = (\gamma p(x)(1 - \frac{\omega}{\alpha}))^{1/(1-\gamma)}$. To have a zoning restriction such that $S_m(x) < S^*(x)$, we only need $\omega > 0$. Intuitively, when local residents have a preference for lower density (captured by $\omega > 0$), they choose zoning that restricts density below the unconstrained developer optimum.

2.1.4 Transit Infrastructure

Suppose a metropolitan transit authority chooses the length of a transit line by solving

$$\max_{X_T} \int_0^{X_T} vx n_m(x) dx - F - cX_T.$$

where v is the value of transit accessibility (the value of unit travel cost saved per trip), and $cX_T + F$ represents the cost of building the transit line. Under the local municipality's zoning restriction, population density along the transit line is $n_m(x) = \frac{p(x)S_m(x)^\gamma}{\alpha}$. Solving the transit authority's problem gives the first-order condition:

The optimal transit length is

$$X_T^* = \frac{1}{\kappa} \log \left(\frac{vN_0}{c} \right),$$

and the building condition is

$$\frac{vN_0}{\kappa} \left(1 - \frac{c}{vN_0} \right) - \frac{c}{\kappa} \log \left(\frac{vN_0}{c} \right) \geq F$$

where $\kappa = \frac{\tau}{\alpha} (1 + \frac{\gamma}{1-\gamma} \frac{\omega}{\alpha})$ captures the effective density cost along the transit line, and $N_0 = \frac{[\gamma(1-\frac{\omega}{\alpha})]^{\frac{1}{1-\gamma}}}{\alpha} p(0)^{\frac{1}{1-\gamma}}$ is the population density at the CBD. The optimal transit length is increasing in the value of transit accessibility, provided $vN_0 > c$ and maximum net surplus is non-negative. Intuitively, N_0 decreases with ω (more restrictive zoning), which leads to lower population density along the transit line, lower returns to transit investment, and discourages transit expansion.

2.2 Quantitative Spatial Model

2.2.1 Setup

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, N\}$. These locations are partitioned into a set of municipalities $g \in G$, such that

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed amount of land $\bar{H}_i$, which can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

Land use is further subject to zoning regulations. Let Z_i^R and Z_i^F denote the residential and commercial zoning restrictions at location i . These regulations impose upper bounds on feasible development:

$$H_i^R \leq \bar{H}_i^R(Z_i), \quad H_i^F \leq \bar{H}_i^F(Z_i),$$

where stricter zoning lowers the feasible density of development.

Locations are connected through a transportation network. For any pair of residence and workplace locations (i, j) , households can choose a commuting route $r \in R_{ij}$. The generalized commuting cost associated with route r is denoted by τ_{ijr} , which depends on transportation infrastructure and congestion:

$$\tau_{ijr} = \tau_{ijr}(I^R, I^T, \mathbf{n}),$$

where I^R denotes metropolitan road investment, I^T denotes metropolitan public transit investment, and $\mathbf{n}$ collects equilibrium traffic and transit usage across routes, which implies congestion force. Road investment primarily lowers the cost of private commuting,

whereas transit investment lowers the cost of public commuting and is more valuable when land around high-access locations can be developed more intensively.

There are three types of agents in the model: households, firms, and governments. Households choose where to live, where to work, and how to commute. Competitive firms choose labor and commercial land to produce a freely tradable good. Government decisions are made at two levels. Municipal governments choose local zoning regulations, while a metropolitan transportation authority chooses transportation investment for the urban area as a whole.

2.2.2 Households

A household chooses a residence location i , a workplace location j , and a commuting route $r \in R_{ij}$. Conditional on these choices, the household consumes a composite tradable good Q_{ij} and residential land h_{ij}^R . Preferences are given by

$$u_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \left(\frac{Q_{ij}}{\beta} \right)^\beta \left(\frac{h_{ij}^R}{1-\beta} \right)^{1-\beta} v_{ijr}(w),$$

where B_i is the residential amenity at location i , $\beta \in (0, 1)$, and $v_{ijr}(w)$ is an idiosyncratic preference shock.

Let q_i^R denote the rental price of residential land at location i , and w_j the wage earned at workplace location j . The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R = w_j.$$

Solving the within-choice consumption problem yields the indirect utility

$$V_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \frac{w_j}{(q_i^R)^{1-\beta}} v_{ijr}(w),$$

up to a constant common to all choices.

To allow for a nested discrete-choice structure, let the idiosyncratic shocks follow a generalized extreme value distribution. The lower nest corresponds to route choice conditional on a residential-workplace pair (i, j) , while the upper nest corresponds to the joint choice of residence and workplace. Formally,

$$G(\{v_{ijr}(w)\}) = \exp \left(- \sum_{i,j \in S^2} \left(\sum_{r \in R_{ij}} v_{ijr}^{-1/\rho} \right)^{\rho/\theta} \right),$$

where ρ and θ govern substitution within and across nests.

Conditional on (i, j) , households choose the commuting route that maximizes indirect utility. At the upper level, they choose residence and workplace locations based on the expected value from the lower nest. Thus, household sorting across space depends on wages, rents, local amenities, commuting costs, and transportation infrastructure.

2.2.3 Firms

At each workplace location j , competitive firms produce a freely tradable good using effective labor and commercial land. Let output be given by

$$Y_j = A_j F(L_j, H_j^F),$$

where A_j is location-specific productivity, L_j is labor employed at location j , and H_j^F is commercial land use. Firms solve

$$\max_{L_j, H_j^F} A_j F(L_j, H_j^F) - w_j L_j - q_j^F H_j^F,$$

where q_j^F denotes the rental price of commercial land.

Profit maximization implies standard labor-demand and land-demand conditions. In equilibrium, wages and commercial rents adjust so that labor and land markets clear.

2.2.4 Municipal Zoning Decisions

Each municipality $g \in G$ chooses a vector of zoning policies

$$Z^g = \{Z_i\}_{i \in S^g}.$$

To avoid modeling an explicit dynamic voting game, I assume municipal governments act as reduced-form representatives of incumbent residents. This captures the idea that local land-use policy is shaped by the interests of existing residents, who value neighborhood amenities, lower congestion, and higher property values, but do not internalize metropolitan-wide effects on housing supply, commuting, or transit efficiency.

Accordingly, municipality g solves

$$\max_{Z^g} W^g(Z^g; I^R, I^T, Z^{-g}),$$

where Z^{-g} denotes the zoning choices of other municipalities, and W^g is incumbent welfare in municipality g . In reduced form, this objective can be written as

$$W^g = \sum_{i \in S^g} [\omega_1 P_i + \omega_2 B_i^{\text{loc}} - \omega_3 N_i],$$

where P_i captures local property-value considerations, B_i^{loc} captures neighborhood amenities, and N_i captures congestion or crowding costs associated with additional development. Because this objective is local, municipalities do not internalize the effect of zoning restrictions on regionwide housing allocation, commuting patterns, or the return to regional transit investment.

2.2.5 Metropolitan Transportation Authority

A metropolitan transportation authority chooses investment in road and transit infrastructure for the region as a whole. Let its policy vector be

$$I = (I^R, I^T).$$

Taking local zoning as given, the authority solves

$$\max_{I^R, I^T} W^M(I^R, I^T; Z) - C^R(I^R) - C^T(I^T),$$

where $W^M(\cdot)$ denotes the regional benefit from transportation infrastructure, Z is the vector of municipal zoning policies, and $C^R(\cdot)$, $C^T(\cdot)$ are the costs of road and transit investment.

The key institutional friction is that the metropolitan authority controls transportation investment but does not control zoning. Since local land use determines whether high-access locations can absorb additional residential and commercial development, the marginal return to transit investment depends on municipal zoning decisions. In particular, restrictive zoning near transit-accessible locations weakens the density and ridership response to public transit investment.

2.2.6 Equilibrium

A decentralized equilibrium consists of

$$\{q_i^R, q_i^F, w_j, H_i^R, H_i^F, L_j, Z^g, I^R, I^T\}_{i,j \in S, g \in G}$$

such that:

1. Households choose residence, workplace, and commuting route to maximize utility.
2. Firms choose labor and commercial land to maximize profits.
3. Each municipality chooses zoning to maximize incumbent resident welfare.
4. The metropolitan transportation authority chooses road and transit investment to maximize regional net benefits, taking zoning as given.
5. Residential land, commercial land, and labor markets clear.

2.2.7 Inefficiency and the Source of Sprawl

The main distortion in the model arises from a mismatch in governance scales. Zoning is chosen locally by municipalities that represent incumbent interests, while transportation is chosen at the metropolitan level. Municipalities have incentives to restrict development in high-access locations in order to preserve local amenities and limit congestion. However, they do not internalize the broader metropolitan costs of such restrictions, including higher housing prices, greater job-housing separation, and lower returns to public transit investment.

As a result, restrictive zoning can prevent transportation improvements—especially transit investments—from translating into higher-density development near well-connected locations. Growth is instead pushed toward lower-cost, more distant areas, contributing to urban sprawl. In this sense, sprawl in the model is not simply a reflection of geography or household preferences, but may emerge as an inefficient equilibrium outcome of fragmented land-use regulation combined with metropolitan transportation investment.

2.2.8 Efficient Benchmark

To evaluate the inefficiency of the decentralized equilibrium, I compare it to a coordinated benchmark in which a planner chooses both transportation investment and zoning:

$$\max_{I^R, I^T, Z} W(I^R, I^T, Z) - C^R(I^R) - C^T(I^T),$$

where $W(\cdot)$ denotes aggregate metropolitan welfare. Unlike individual municipalities, the planner internalizes the effect of land-use restrictions on housing supply, commuting, and the productivity of transportation infrastructure.

Comparing the decentralized equilibrium with this benchmark highlights the extent to which fragmented zoning lowers the return to transit, biases investment toward road-oriented development, and generates excessive urban sprawl.

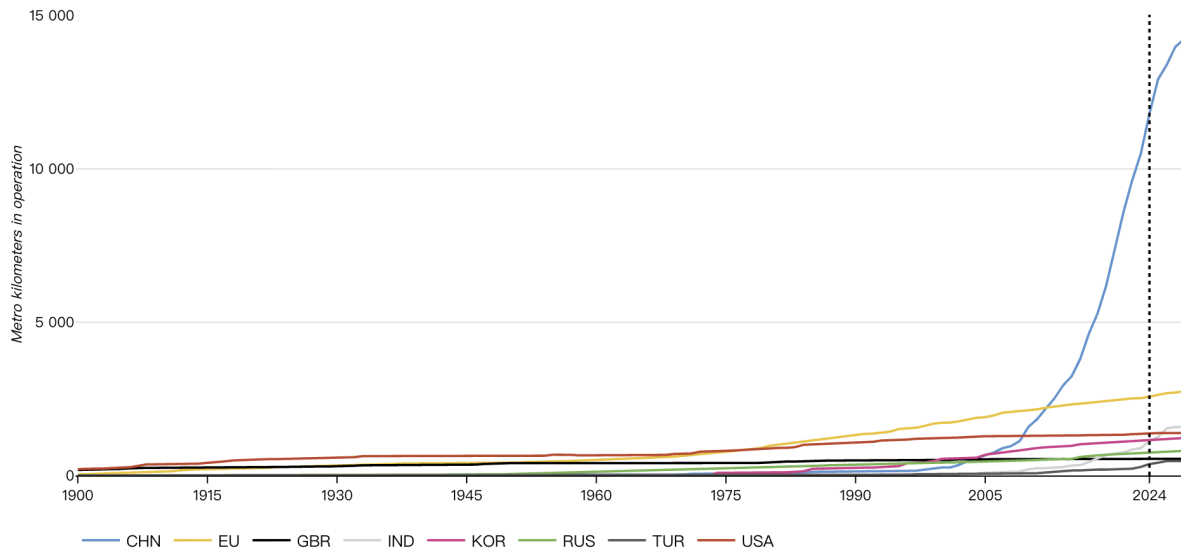


Figure 2: Metro kilometers in operation

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